

Solving Quadratics by Factoring

Name: _____

Algebra I | Guided notes — we fill these in together

Vocabulary

Zero Product Property: if $A \cdot B = \underline{\hspace{1cm}}$, then $A = \underline{\hspace{1cm}}$ or $B = \underline{\hspace{1cm}}$.

Root (solution): an x that makes the equation $\underline{\hspace{2cm}}$; on a graph it is an $\underline{\hspace{2cm}}$.

We do together. Solve $x^2 + 5x + 6 = 0$.

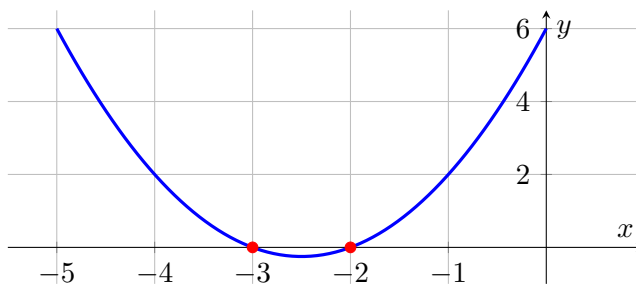
Step 1. One side already = 0. Good.

Step 2. Factor: $x^2 + 5x + 6 = (\underline{\hspace{1cm}})(\underline{\hspace{1cm}})$

Step 3. Set each factor = 0: $\underline{\hspace{1cm}} = 0$ or $\underline{\hspace{1cm}} = 0$

Step 4. Solve: $x = \underline{\hspace{1cm}}$ or $x = \underline{\hspace{1cm}}$

See it: the roots are where the parabola crosses the x -axis.



You Try. Solve each. Show every step, then name the roots.

a) $x^2 - x - 12 = 0$

b) $x^2 - 9 = 0$

Summary (fill in): To solve a quadratic by factoring: make one side = $\underline{\hspace{1cm}}$, $\underline{\hspace{1cm}}$ the other side, then use the $\underline{\hspace{2cm}}$ and solve each factor.